

Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure

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PUBLICATION YEAR	1976
JOURNAL	Journal of Financial Economics, vol. 3, no. 4, pp. 305–360
DOI	10.1016/0304-405X(76)90026-X
RESEARCH AREA	Corporate Finance — Theory of the Firm, Agency Theory, Capital Structure
SERIES	Corporate Finance
PAPER TYPE	Theoretical (positive economic theory; no formal statistical estimation)
REVIEWER	Arthur Faugeron
DATE REVIEWED	20 August 2026

ABSTRACT

Jensen and Meckling ask why the ownership structure of the firm — the division of claims among an owner-manager, outside shareholders, and debtholders — takes the particular forms observed historically, given that separating ownership from control is costly. Building on the property-rights and agency-relationship literatures, they model the manager as a utility maximizer who values both wealth and non-pecuniary perquisites, and define agency costs as the sum of monitoring expenditures, bonding expenditures, and the residual loss that remains even after optimal monitoring and bonding. Selling outside equity dilutes the manager's claim on firm value and induces him to consume more perquisites and to under-invest relative to the value-maximizing scale; selling debt creates an incentive to shift toward riskier projects at the bondholders' expense and gives rise to monitoring, bonding, and bankruptcy costs. The paper's central claim is that the firm's ownership structure is the outcome of the manager minimizing the sum of these two cost schedules, subject to his personal wealth constraint and the economy's demand for outside financing. The analysis yields a rich set of qualitative predictions about corporate financing choices, all derived from optimizing behavior under positive, not zero, contracting costs.

1 Introduction

Jensen and Meckling (1976) is one of the most cited papers in economics, and its influence on the theory of corporate finance is difficult to overstate: it supplied the conceptual vocabulary — agency costs, monitoring, bonding, residual loss, the firm as a nexus of contracts — that the subsequent forty years of corporate-finance and corporate-governance research still uses. The paper does not report an empirical test, estimate a regression, or calibrate a model to data. It is a piece of positive economic theory: a formal argument, built from utility maximization and a diagrammatic apparatus, about why the ownership structure of the firm looks the way it does when contracting between an entrepreneur-manager and outside financiers is costly rather than free.

1.1 What is the paper trying to discover?

The paper asks a question that the neoclassical theory of the firm, as the authors characterize it, had no apparatus to answer: given that a firm can be financed by some mixture of the manager's own equity, outside equity sold to passive shareholders, and debt sold to bondholders, what determines the mixture actually chosen, and why does that mixture generate a firm value below the value that would obtain under costless contracting? The authors are explicit that this is a question about the *relative amounts* of insider versus outsider claims on the firm, not merely about the debt-to-equity ratio in the conventional capital-structure sense.

1.2 Why does the question matter?

The question matters because the standard theory of the firm, in the authors' description, treats the firm as a "black box" that maximizes profit by construction, with no mechanism explaining how the conflicting objectives of the individuals who compose the firm are reconciled into that outcome. Contemporary debates over the "separation of ownership and control," the social responsibility of the corporation, and the determinants of capital structure all presupposed answers to questions the existing theory could not derive. A theory that explains ownership structure as the equilibrium outcome of self-interested contracting, rather than as an exogenous institutional fact, replaces normative assertions about managerial discretion with testable, comparative-static predictions.

1.3 What was known before?

Three literatures existed in parallel without being joined. The property-rights literature associated with Coase (1937, 1960) and Alchian and Demsetz (1972) had established that the specification of rights in a contract determines the incentives of the parties to it, and that the firm is best understood as a set of contracting relationships rather than as a locus of authority. The theory of agency, developing contemporaneously through the work of Ross (1973) and Wilson (1968), had formalized the problem of inducing an agent to act in a principal's interest under costly monitoring, but had concentrated on the normative design of incentive contracts. The theory of

finance possessed the Modigliani-Miller (1958, 1963) irrelevance propositions and their tax-adjusted extensions, together with the nascent Black-Scholes-Merton option-pricing framework, but as the authors document by quoting Fama and Miller (1972, p. 173) directly, finance theory had no account of what actually determines the debt-equity mix a firm chooses.

1.4 What is missing from the literature?

What was missing was a theory connecting the three literatures: an explanation of the ownership structure of the firm that treats debt and outside equity as claims whose issuance changes the incentives facing the manager, and that derives an equilibrium mixture of insider and outsider claims from those incentive effects rather than assuming it. In particular, no existing framework explained why an entrepreneur would voluntarily sell equity in his own firm even though doing so provably reduces the firm's value, why debt was used long before it carried any tax advantage, or why highly regulated firms such as utilities and banks carry more debt than unregulated firms of comparable risk.

1.5 What does the paper propose?

The paper proposes to model the manager as maximizing his own utility, defined over wealth and non-pecuniary benefits, subject to a set of budget constraints determined by how much of the firm he has sold to outsiders. Selling outside equity and selling debt each change these constraints in a distinct, formally tractable way, and each generates its own schedule of agency costs. The manager, who bears the full capitalized cost of both schedules because rational outside claimants price the anticipated agency costs into what they will pay, chooses the mixture of inside equity, outside equity, and debt that minimizes their sum.

1.6 How does it investigate the question?

The method is comparative-statics diagrammatic and algebraic analysis under an explicit set of simplifying assumptions (no taxes, no trade credit, single-period, single decision-maker), built up in stages: first the agency costs of outside equity alone, holding debt at zero; then the agency costs of debt alone, holding outside equity at zero; then a synthesis in which both instruments are available simultaneously. There is no dataset, no estimation, and no statistical test anywhere in the paper; the empirical content of the analysis is the list of qualitative comparative-static predictions it generates, several of which the authors state explicitly as historical or cross-sectional regularities the theory is intended to rationalize.

1.7 What does it find?

The central finding is that a value-maximizing entrepreneur, facing a financing need larger than his personal wealth, will not maximize the market value of the firm; the wedge between the value that would obtain under 100 percent ownership (or costless contracting) and the value actually realized is the total agency cost of the ownership structure the manager rationally chooses. This wedge is positive whenever monitoring is costly, is independent of whether the firm faces product-market competition or monopoly power, and is minimized — not eliminated — by an interior

mixture of debt and outside equity that trades the perquisite-consumption and under-investment costs of outside equity against the risk-shifting, monitoring, bonding, and bankruptcy costs of debt.

1.8 Why does the finding matter?

The finding reframes a reduction in firm value below the frictionless benchmark as an efficient, not a pathological, outcome: agency costs are the price of obtaining outside capital under positive contracting costs, exactly as the cost of iron ore is the price of a scarce input, and comparing a world with agency costs to a hypothetical frictionless world and calling the former "inefficient" is, in the authors' own framing borrowing from Demsetz (1969), a Nirvana fallacy. This reframing licenses a large research program: because agency costs are priced into the securities issued, they should show up systematically in the cross-section of financing choices, covenant design, ownership concentration, and executive compensation — a program the corporate-finance and corporate-governance literatures pursued for decades after 1976.

2 Research Problem and Hypothesis

2.1 Research Question

Stated precisely: given an entrepreneur-manager with personal wealth W smaller than the investment outlay I required to exploit a profitable investment opportunity, and given that the only claims that can be issued are non-voting outside equity and limited-liability debt, what combination of inside equity, outside equity, and debt minimizes the sum of (i) the agency costs generated by the manager's incentive to consume non-pecuniary perquisites and under-invest once his fractional claim on firm value falls below one, and (ii) the agency costs generated by the manager's incentive to shift toward higher-variance, lower-expected-value projects once fixed debt claims are outstanding? The question is explicitly positive, not normative: the paper does not ask how a social planner should structure ownership, but what structure self-interested contracting between rational parties actually produces.

2.2 Motivation

The motivation is that the "separation of ownership and control," a phrase inherited from Berle and Means (1932) and traceable to Adam Smith's 1776 remark — quoted as the paper's epigraph — that the directors of joint-stock companies cannot be expected to watch over other people's money with the vigilance of a private partner, had generated a large popular and professional literature asserting that managers pursue their own interests at shareholders' expense, without a corresponding theory of why shareholders would rationally tolerate this, why the corporate form nonetheless proliferated, or what specific contractual and financial mechanisms limit the resulting losses. A reader working in corporate finance, governance, or contract theory should

engage with this paper because it is the direct ancestor of the free cash flow theory (Jensen, 1986), the debt-covenant and bonding literatures, and much of empirical corporate-governance research on ownership concentration and managerial incentives.

2.3 Existing Literature

PAPER. The authors position their contribution against three specific bodies of prior work, all named directly in the introduction. Coase (1937) is credited with posing the question of the boundary of the firm but, in Jensen and Meckling's reading, without a positive theory of what determines that boundary beyond a comparison of the cost of using markets against the cost of using authority. Alchian and Demsetz (1972) are credited with correctly emphasizing that the firm is a nexus of voluntary contracts rather than a locus of authority, and with formalizing the monitoring problem in joint-input, team-production settings — but criticized for restricting the source of agency costs too narrowly to joint production, when Jensen and Meckling argue agency costs arise in any contractual relationship, joint production or not. Modigliani and Miller (1958, 1963) had shown that in the absence of bankruptcy costs and tax subsidies, firm value is independent of capital structure, and later, once a tax subsidy on interest is admitted, that firm value should rise monotonically with leverage — a prediction inconsistent with the moderate, historically stable leverage ratios actually observed, a tension Modigliani and Miller themselves acknowledged without resolving (their appeal to "the need for preserving flexibility" is quoted directly in the paper).

2.4 Research Gap

The gap, stated in the authors' own terms, is that no existing theory explained the fraction of equity claims held by insiders versus outsiders, that bankruptcy-cost theories of capital structure implied zero debt in the absence of a tax subsidy (counterfactually, since debt predates the relevant tax subsidies), and that neither tax subsidies nor bankruptcy costs could explain the use of preferred stock or warrants, which carry no comparable tax advantage. The gap is therefore not "a theory of capital structure is missing" in a generic sense — Modigliani-Miller and its tax-adjusted successors already constituted such a theory — but that the existing theories could not jointly explain the observed persistence of moderate leverage, the coexistence of debt and outside equity, and the specific contractual features (covenants, audited financial statements, voluntary disclosure) surrounding both.

2.5 Contribution

PAPER. The claimed contribution is a positive theory in which ownership structure is derived, not assumed: the manager's incentive to consume perquisites once equity is sold, and his incentive to shift risk once debt is sold, are shown to generate value losses that outside claimants rationally anticipate and price, so that the manager alone bears the capitalized cost of his own future opportunism. The theory is claimed to explain, among other things, why debt was used before its tax advantage existed, why regulated utilities and banks carry more debt than equivalent-risk unregulated firms, and why security analysis is socially useful even though average investors cannot beat the market on a risk-adjusted basis.

CRITICAL EVALUATION OF THE CONTRIBUTION

CRITIQUE. The claimed contribution is genuinely novel in its synthesis, but several of the ten numbered implications listed in the introduction (Section 1 of the original paper) are asserted rather than formally derived within the body of the paper as published; the analysis of Sections 2 and 4 rigorously derives the agency-cost-of-equity and agency-cost-of-debt schedules, but the specific claims about preferred stock, voluntary auditing, and regulated-industry leverage are sketched narratively rather than proved as propositions. This is a difference between what the paper formally establishes and what it informally motivates, addressed further in Section 7 below.

3 Theoretical Framework and Mechanism

3.1 The definition of agency costs

The paper's foundational definition, on which the entire analysis rests, is that an agency relationship is a contract under which a principal engages an agent to perform a service involving delegated decision-making authority, and that whenever both parties are utility maximizers there is reason to expect the agent will not always act in the principal's interest. Agency costs are defined as the sum of three components: the monitoring expenditures by the principal, the bonding expenditures by the agent, and the residual loss — the dollar-equivalent reduction in the principal's welfare from the remaining divergence between the agent's actual decisions and the decisions that would maximize the principal's welfare, given optimal monitoring and bonding. This tripartite decomposition is the paper's central analytical device and recurs in both the equity and the debt analysis.

3.2 The agency costs of outside equity

Consider an owner-manager who derives utility from both pecuniary wealth and a vector $X = \{x_1, x_2, \dots, x_n\}$ of non-pecuniary benefits — office space, staff amenities, favored suppliers, and similar perquisites — consumed inside the firm. Let $C(X)$ be the dollar cost of providing X , $P(X)$ its dollar value to the firm's output, and $B(X) = P(X) - C(X)$ the net benefit to the firm, ignoring effects on the manager's wage. The value-maximizing level X^* of perquisites is defined by the first-order condition:

X^* the vector of factor and activity levels satisfying

$$\partial P(X^*)/\partial X^* = \partial C(X^*)/\partial X^* = 0 \quad (1)$$

Let F denote the current market value of the stream of the manager's non-pecuniary consumption. Because one dollar of F withdrawn reduces firm value by exactly one dollar, the 100-percent-owner's opportunity set between firm value V and F is a line of slope -1 ; his optimum, at the tangency with his indifference curve between wealth and perquisites, defines V^* and F^* . If he sells

a fraction $(1-\alpha)$ of the equity and retains α , the cost to him of consuming one dollar of perquisites falls from one dollar to α dollars, so his personal opportunity locus flattens to slope $-\alpha$, and he rationally increases consumption of F beyond F^* .

THE PRICING THEOREM

The paper proves, by contradiction, that a rational outside buyer paying S_0 for a claim $(1-\alpha)$ on the firm will price the manager's anticipated increase in perquisite consumption into the price paid, so that the tangency of the manager's new opportunity locus with his indifference curve occurs exactly on the original value-frontier VF . Writing the manager's total post-sale wealth as

$$W = S_0 + S_i = S_0 + \alpha V(F, \alpha) \quad (2)$$

where S_i is the value of the manager's retained stake, the proof shows that the equilibrium price satisfies $(1-\alpha)V' = S_0$, which implies $W = V'$: the entire decline in firm value, $V^* - V'$, is borne by the manager alone, not shared with the outside buyer. This is the paper's key distributional result and the reason the manager, not a diffuse and passive shareholder base, has the incentive to minimize agency costs: he captures one hundred percent of any reduction in them.

OPTIMAL SCALE OF THE FIRM UNDER OUTSIDE EQUITY

Allowing the level of investment I to vary, and defining I^* as the value-maximizing, zero-agency-cost scale at which $\Delta V - \Delta I = 0$, the manager's actual optimum I' under outside financing satisfies:

$$\Delta V - \Delta I + \alpha' \Delta F = 0 \quad (3)$$

which, substituting $V = V^* - F$ (where V^* is firm value at zero perquisite consumption), rearranges to

$$(\Delta V^* - \Delta I) - (1-\alpha') \Delta F = 0 \quad (4)$$

Since $\Delta F > 0$ whenever any outside equity is sold, condition (4) requires $(\Delta V^* - \Delta I) > 0$ at the chosen scale I' ; because the net-value function $V(I) - I$ is concave with its maximum at I^* , a strictly positive marginal net value at I' implies $I' < I^*$. This is a genuinely non-obvious prediction: the model does not merely say the manager over-consumes perquisites at a given scale, it says he stops expanding the firm *before* reaching the value-maximizing scale, because further expansion would require selling more outside equity and thereby incurring more perquisite-consumption losses at the margin than the value the expansion would create net of that loss. Formally, allowing costly monitoring (a function $F(M, \alpha)$ with $\partial F / \partial M < 0$ and $\partial^2 F / \partial M^2 > 0$) and bonding expenditures b pushes the realized value back toward $V = V(I) - F(M, \alpha) - M$, and the manager chooses M and b up to the point where their marginal cost equals their marginal contribution to his wealth — but monitoring and bonding never drive gross agency costs to zero as long as either activity has a strictly positive marginal cost.

3.3 The agency costs of debt

The debt analysis begins from a deliberately stylized setting: two mutually exclusive, equal-cost investment projects with log-normally distributed terminal payoffs x_1 and x_2 , identical expected value $E(\bar{x})$, identical systematic (covariance) risk in the Sharpe-Lintner CAPM sense, but different variances $\sigma^2_1 < \sigma^2_2$. Because expected payoff and systematic risk are equal, CAPM implies the two projects have identical unlevered firm value V . Let the firm issue a non-coupon bond promising face value X^* , so bondholders receive

$$R_j = X^* \text{ if } x_j \geq X^*; R_j = x_j \text{ if } x_j < X^* \quad (5)$$

Jensen and Meckling then invoke the Black and Scholes (1973) result directly — citing it by name — that levered equity is formally a European call option on total firm value with strike price X^* : the residual claimants have, in effect, the right to buy the firm back from the bondholders for X^* at maturity. Since option value is strictly increasing in the variance of the underlying, and the two projects share the same underlying value V but differ in variance, equity value satisfies $S_2 > S_1$, and since $B_j = V - S_j$ for each project j , bond value satisfies $B_1 > B_2$. A manager who sells debt priced as if the firm will take the low-variance project, then switches to the high-variance project, transfers wealth from bondholders to himself as residual claimant — the asset-substitution or risk-shifting problem, derived here roughly a decade before it acquired that standard name in the corporate-finance literature.

WHEN RISK-SHIFTING DESTROYS VALUE

The more damaging case is where the higher-variance project also has strictly lower expected value, so total firm value falls, $V_1 > V_2$. Decomposing the change in value:

$$\Delta V = V_1 - V_2 = (S_1 - S_2) + (B_1 - B_2) \quad (6)$$

the paper shows algebraically that $S_1 - S_2$ can be negative — equity value *rises* with the switch — even though total firm value falls, because the transfer from bondholders, $B_1 - B_2$, exceeds the destruction in total value. This is the paper's sharpest result on debt: absent monitoring, the manager's private incentive can favor a project that destroys value overall, purely because the loss falls disproportionately on fixed claimants. Crucially, the paper notes that with rational, unbiased bondholders this wealth transfer does not actually occur in equilibrium — bondholders who correctly anticipate the switch price the bond at B_2 from the outset — so the true cost borne by the manager is not a realized transfer but the residual loss $V_1 - V_2$ itself, plus whatever monitoring (covenants) and bonding (voluntary reporting, audited statements) expenditures are needed to make the low-variance promise credible.

3.4 Bankruptcy and reorganization costs

PAPER. A third component of the agency cost of debt is added: because writing debt contracts that unambiguously delineate creditor rights under every contingency is itself costly, and because the bankruptcy process consumes real resources in adjudicating claims, the mere possibility of bankruptcy — not only its incidence — reduces the price rational bondholders will pay for fixed claims, even holding the incentive-conflict problem constant.

3.5 Synthesis: the joint determination of ownership structure

INFERENCE / SECONDARY-SOURCE CORROBORATED. The paper's Section 5 combines the two schedules developed above into a single optimization. The agency cost of outside equity rises with the fraction of equity sold to outsiders (the manager's incentive to consume perquisites and under-invest grows as his retained stake α falls); the agency cost of debt rises with the fraction of external financing raised as debt (the risk-shifting incentive, and the monitoring, bonding, and bankruptcy costs it induces, grow with leverage). For a fixed amount of outside financing $K = I - W$ that must be raised from some combination of outside equity S and debt D with $D+S=K$, the manager chooses the split that minimizes the sum of the two agency-cost schedules; because one schedule is rising and the other falling in the debt share $D/(D+S)$, the sum is generically U-shaped and minimized at an interior point $D^*/(D^*+S^*)$, which determines the equilibrium ownership structure. This description is not a verbatim transcription of the primary text — the full text of Section 5 (pp. 342–352 of the published article) could not be retrieved verbatim during the preparation of this review, a limitation disclosed explicitly in Section 12 below — but the qualitative structure summarized here is corroborated by independent secondary treatment of the paper (Gao, 2019, International Core Journal of Engineering, summarizing the same synthesis) and is consistent with every subsequent textbook treatment of the paper's capital-structure implications. It should be read as a reconstruction of the paper's argument, not as a directly quoted derivation.

4 Method and the Nature of the Paper's Empirical Content

This paper contains no data, no sample, no regression, and no statistical test; there is accordingly no empirical design, identification strategy, or estimation method to describe in the conventional sense, and this review does not force such a section. The paper's method is axiomatic and diagrammatic: it specifies a utility function for the manager over wealth and non-pecuniary benefits, a small number of permanent and temporary simplifying assumptions (zero taxes, no trade credit, non-voting outside equity, no complex claims such as convertibles, a single production-financing decision, a single manager), and derives comparative-statics results from constrained optimization, using indifference-curve diagrams (their Figures 1 through 4) as the primary expository device.

The "empirical content" of the paper, in the sense relevant to a literature review, is the set of qualitative, falsifiable predictions it generates — listed by the authors themselves as ten numbered implications in the introduction, including that debt was used as a financing source before it

carried any tax advantage, that highly regulated industries such as public utilities and banks should carry higher debt-equity ratios than unregulated firms of comparable risk, and that owner-managed firms whose sole external financing is borrowing should be concentrated in particular industries. None of these ten claims is tested against data anywhere in the paper; they are offered as predictions the theory is *capable of* rationalizing, to be tested by future empirical work, and the paper should be read and cited accordingly — as the origin of an empirical research program, not as an empirical paper itself.

5 Results: Predictions and Table/Figure Analysis

5.1 What the model actually delivers

Table 1 summarizes the paper's four principal diagrams (its Figures 1–4), each of which is a formal comparative-statics device rather than a decorative illustration, and states what each one establishes.

Table 1. The paper's diagrammatic apparatus and what each figure establishes

Figure	Axes / construction	What it establishes
Fig. 1	Firm value V (vertical) against market value of manager's non-pecuniary benefits F (horizontal); indifference curves U_1, U_2, U_3 over wealth and perquisites	Derives the pricing theorem: the entire decline in firm value from V^* to V' after a partial equity sale is borne by the manager, since rational buyers price the anticipated perquisite increase.
Fig. 2	Post-investment wealth against F , for varying investment scale I , comparing the 100%-ownership expansion path to the fractional-ownership path	Shows the manager's chosen scale I' falls short of the value-maximizing scale I^* once outside equity is required, and defines gross agency cost A as the resulting value gap.
Fig. 3	Same axes as Fig. 1, adding budget locus BCE generated by costly monitoring, $F(M, \alpha)$	Shows that positive but finite monitoring expenditure M narrows, but does not eliminate, the perquisite-driven value loss; establishes the interior optimum for M .
Fig. 4	Expansion paths under 100% ownership, zero monitoring, and optimal monitoring/bonding, overlaid	Combines Figs. 2 and 3 to characterize the fully optimized scale, optimal monitoring level M , and optimal bonding level b , simultaneously, at point G .

Source: reconstructed by the reviewer from the diagrammatic descriptions and accompanying algebra in Jensen and Meckling (1976, pp. 316–327); not a reproduction of the original artwork.

5.2 The ten stated implications

The introduction lists ten phenomena the theory purports to explain. Economically, the most important are: (i) a mixed-financing entrepreneur chooses a lower firm value than a sole owner would, regardless of whether the firm is a monopolist or operates in a competitive product market, because the agency-cost mechanism is a contracting friction internal to the firm and unrelated to product-market structure; (ii) debt was used as a financing source before it carried any interest-

tax-shield advantage, which the theory rationalizes through the debt-agency-cost tradeoff rather than through taxation; and (iii) highly regulated industries such as public utilities and banks should exhibit higher debt-equity ratios than unregulated firms of comparable risk, because regulatory oversight substitutes for some of the monitoring that bondholders would otherwise have to purchase directly, lowering the marginal agency cost of debt for a given leverage ratio.

ECONOMIC SIGNIFICANCE, NOT MERELY STATISTICAL SIGNIFICANCE

Because the paper contains no data, "economic significance" here means something different from the usual empirical-paper sense: it means that the mechanisms identified (perquisite consumption, under-investment, risk-shifting, monitoring and bonding, bankruptcy costs) are shown, by explicit construction in Figs. 1–4 and equations (1)–(6) above, to be first-order in sign and direction, not that any coefficient has been estimated to be economically large. Whether the mechanisms are quantitatively large in real firms is a question the paper leaves entirely to future empirical work, and the reviewer is not aware of any calibration exercise within the paper itself that would support a claim of large or small magnitude.

6 Robustness and Additional Analysis

Because there is no empirical estimation, there are no robustness checks in the conventional sense (alternative samples, alternative specifications, placebo tests). The paper instead subjects its own qualitative claims to several internal consistency checks, each evaluated below for whether it actually resolves the concern it addresses.

6.1 Monopoly versus competition invariance

PAPER. Concern: does the existence of product-market competition eliminate agency costs by forcing managers who consume excessive perquisites out of business, as Friedman (1970) had argued? **EVIDENCE.** The authors show the manager equates the marginal cost of monitoring to the marginal wealth gain from reduced perquisite consumption regardless of the degree of product-market power, because the mechanism operates through the manager's personal budget constraint, not through the firm's profit function directly. **CRITIQUE.** This does resolve the specific claim that competition mechanically drives agency costs to zero, but it does not address a weaker and more plausible version of the competitive-discipline argument — that competition, by increasing the probability a poorly monitored firm is acquired or a mismanaging executive is replaced, lowers the *equilibrium level* of monitoring costs required to sustain a given level of agency costs, an argument the paper acknowledges only briefly in its discussion of the market for the firm as a disciplining device.

6.2 Limited liability as a sufficient explanation

PAPER. Concern: is limited liability, on its own (following Manne, 1967 and Alchian and Demsetz, 1972), sufficient to explain reliance on diffuse outside equity? **EVIDENCE.** The authors point out that ordinary debt also carries limited liability, so limited liability cannot by itself explain why we do

not observe large, individually owned corporations financed almost entirely by borrowing. **CRITIQUE.** This is a genuine and still-cited objection; it correctly identifies limited liability as necessary but not sufficient. It does not, however, fully quantify how much of the observed reliance on outside equity is explained by the agency-cost mechanism versus other frictions (information asymmetry, adverse selection in Myers-Majluf's 1984 sense) that postdate this paper and were not available to the authors.

6.3 Comparison with the Modigliani-Miller irrelevance benchmark

PAPER. Concern: does the Modigliani-Miller (1958, 1963) irrelevance theorem, once extended for bankruptcy costs and tax subsidies, already provide an adequate theory of capital structure? **EVIDENCE.** The authors note that the bankruptcy-cost-plus-tax-subsidy theory implies zero debt whenever no tax subsidy exists, which is inconsistent with the historical use of debt prior to the relevant tax provisions. **CRITIQUE.** This is a correct and important empirical observation, but the paper does not itself supply a dated series of corporate leverage ratios or tax-code changes to substantiate the claim quantitatively; the assertion rests on general historical knowledge rather than a documented data series, and a rigorous test of this specific claim was left to the subsequent capital-structure literature (for example, Graham, 2000, on the value of the debt tax shield).

7 Critical Evaluation: What the Paper Actually Establishes

ESTABLISHED

The pricing theorem (Section 3.2 above): under the paper's assumptions, a rational, competitive equity market prices anticipated managerial opportunism into the sale price of outside equity, so the resulting value loss is borne entirely by the selling owner-manager, not shared with or imposed on the buyer. This is formally proved by contradiction within the paper and does not depend on auxiliary secondary-source corroboration.

ESTABLISHED

The under-investment result (equation 4 above): the manager's chosen investment scale under outside-equity financing is strictly below the value-maximizing scale whenever any positive amount of outside equity is used and perquisite consumption is strictly positive at the margin. This follows algebraically from the concavity of the net-value function and is a clean, if narrow, formal result.

ESTABLISHED

The risk-shifting decomposition (equation 6 above): equity value can rise from a switch to a lower-expected-value, higher-variance project, with the gain financed entirely by a transfer from bondholders, is a valid algebraic decomposition of the change in firm value under the paper's option-pricing mapping (subject to the caveat about the internal inconsistency of

applying Black-Scholes when V itself depends on the debt-equity mix, which the authors themselves flag in a footnote).

SUPPORTED BUT CONDITIONAL

The claim that monitoring and bonding costs, once optimally chosen, restore efficiency "given the agency relationship" (Section 2.5 of the original) is conditional on the specific functional-form assumptions about $F(M, \alpha)$ (diminishing returns to monitoring) and on the absence of any ambiguity in the contractual specification of rights; the authors flag this ambiguity themselves as a potential additional source of agency cost they do not model.

SUGGESTIVE

The ten numbered implications in the introduction — regulated-industry leverage, the historical pre-tax use of debt, the rationale for voluntary auditing — are consistent with the mechanisms formally derived in Sections 2 and 4, but are not themselves formally derived as propositions within the paper; they are narrative extrapolations from the core model, offered as testable predictions rather than established results.

NOT ESTABLISHED

Any claim about the *quantitative* magnitude of agency costs in real firms, any specific numerical optimal debt-equity ratio, and any claim that the synthesis in Section 5 (the joint minimization of the two agency-cost schedules) has a unique, generically interior solution under general functional forms — the paper's own treatment is diagrammatic rather than a general proof of existence and uniqueness, a gap later formalized and qualified by Kraus and Litzenberger (1973) and, critically, by Kürsten (1995), who shows the two cost schedules cannot in general be summed independently because a change in leverage changes the marginal agency cost of equity itself.

7.1 Strengths

PAPER (SELF-IDENTIFIED). The authors themselves emphasize the breadth of the synthesis across three previously separate literatures, and the fact that the theory generates a definition of the firm (a nexus of contracts) with no dependence on treating the firm as an individual decision-maker with its own "objective function."

INFERENCE (INDEPENDENTLY IDENTIFIED). Three further strengths are, in the reviewer's independent judgment, at least as important as those the authors highlight. First, the pricing theorem's distributional result — that the manager bears the full capitalized cost of his own future opportunism — is what makes the subsequent claim that managers have incentives to *minimize* agency costs internally consistent; without it, the theory would need an external enforcement mechanism. Second, the use of the nascent Black-Scholes-Merton option-pricing apparatus to formalize the risk-shifting incentive is methodologically ahead of its time and directly seeded the subsequent options-based literature on corporate liabilities (Smith, 1976; Galai and Masulis, 1976). Third, the tripartite decomposition of agency costs into monitoring, bonding, and residual loss

gives the concept empirical traction: each component is, in principle, separately measurable, which is precisely what later empirical corporate-governance research (board monitoring intensity, covenant tightness, executive equity grants) exploited.

7.2 Weaknesses and Independent Critique

PROBLEM → WHY IT MATTERS → BIAS → HOW IT COULD BE TESTED

Static, single-decision framework. The model explicitly assumes a single production-financing decision with no repeated interaction (assumption P.6). This matters because reputation effects — a manager's incentive to preserve his ability to raise capital in future periods — are a first-order determinant of bonding behavior in practice and are assumed away entirely, likely overstating the residual loss relative to a repeated-game setting. This could be tested by comparing agency-cost proxies (e.g., perquisite consumption, leverage) across young, single-financing-round firms versus seasoned issuers with established reputations.

Independence of the two agency-cost schedules. The Section 5 synthesis treats the agency cost of debt and the agency cost of equity as additively separable functions of the debt and equity shares respectively. Kürsten (1995) shows formally that this separability fails once the manager's ownership stake and the debt-equity ratio interact through the risk-shifting incentive itself — a higher retained equity stake changes the manager's risk-shifting incentive under debt, so the two schedules cannot be minimized independently and then summed. This matters because the paper's central comparative-static prediction (an interior, well-defined optimal leverage ratio) is weaker than presented if the schedules interact.

No treatment of free cash flow absent new financing. The framework as presented models agency costs only in the context of raising new outside capital; it does not address the agency costs of cash retained inside a mature firm with no immediate financing need, a gap Jensen himself closed a decade later (Jensen, 1986) and which Chu (2011) and Wang (2010) argue is a first-order omission for firms generating large internal cash flows.

Risk aversion assumed away. Temporary assumption T.4 ignores portfolio-diversification considerations induced by uncertainty; Gorman (1986) and subsequent commentary note that once the manager is risk-averse and under-diversified (as an owner-manager with concentrated firm-specific wealth typically is), both the debt and equity agency-cost schedules shift, and the manager's optimal retained stake is generally lower than the model without risk aversion would predict, for portfolio reasons entirely independent of agency costs.

Of these, the single most important weakness, in the reviewer's judgment, is the additive-separability assumption underlying the Section 5 synthesis: it is the assumption doing the most work in generating the paper's headline capital-structure prediction (an interior, uniquely determined optimal debt-equity mix), and it is also the assumption shown by the subsequent literature to be least likely to hold generally.

8 Contradictions and Edge Cases

EXTENSION. Several extreme-value and boundary checks, none performed explicitly in the paper, are useful for probing where the mechanism breaks down.

What happens as $\alpha \rightarrow 1$? As the manager's retained ownership share approaches one, the agency cost of outside equity in equation (4) vanishes by construction, since $(1-\alpha)\Delta F \rightarrow 0$; the model correctly nests the 100-percent-owner benchmark as a limiting case, which is a basic but necessary internal-consistency check the paper passes.

What happens as $\alpha \rightarrow 0$? As the manager's stake approaches zero, condition (4) implies the manager places essentially no weight on firm value at the margin, so perquisite consumption should rise to the point where the marginal utility of an additional dollar of perquisites is driven only by his (near-zero) share of the cost — predicting near-total capture of firm resources for private benefit. Real diffuse-ownership public corporations, where individual insider stakes can be a small fraction of one percent, do not exhibit total capture; this is consistent with the model only once monitoring by boards, analysts, and the market for corporate control (mechanisms the paper gestures at in Section 2.6 but does not formally embed in the same diagram) is added, suggesting the two-variable (F, α) framework alone is insufficient at this boundary and the paper's own qualitative appeal to external monitoring mechanisms is doing unmodeled work.

Does the result survive if bondholders are as well informed as the manager? The risk-shifting result (equation 6) depends on bondholders being unable to prevent the switch *ex post*; if bondholders can costlessly observe and enforce the promised project choice, no transfer occurs and no residual loss arises — the entire debt-agency-cost result collapses to zero, which is consistent with the paper's own claim that monitoring costs, not the mere existence of debt, are the source of the cost. This is a reassuring internal consistency check: the theory correctly predicts zero agency cost of debt in the frictionless-information limit.

Could another mechanism generate the same predictions? The under-investment result (Section 3.2) is observationally similar to the debt-overhang under-investment result of Myers (1977), cited by Jensen and Meckling themselves in a footnote, which arises from an entirely different mechanism (existing debt overhang deterring new positive-NPV investment because gains would accrue to bondholders). A researcher observing under-investment in a levered firm cannot, from this paper alone, distinguish the Jensen-Meckling perquisite-driven equity-agency-cost channel from the Myers debt-overhang channel without additional identifying variation, since both predict the same qualitative sign.

9 Further Literature

PRECEDED IT

Coase (1937), “The Nature of the Firm.” Shared idea: the firm's boundary and internal organization are determined by contracting costs, not technology alone. Relationship: Jensen and Meckling adopt Coase's contractual view of the firm but reject his emphasis on authority as the organizing principle, substituting the agency-cost-minimizing nexus of contracts. It matters because it is the direct conceptual ancestor of the paper's central definitional move.

Alchian and Demsetz (1972), “Production, Information Costs, and Economic Organization.” Shared idea: monitoring costs in team production explain the existence of a residual claimant. Relationship: Jensen and Meckling generalize the Alchian-Demsetz monitoring problem beyond joint-input team production to any contractual relationship, including the purely bilateral owner-manager/outside-financier relationship that has no joint-input structure at all. It matters because it shows the scope of the agency-cost concept is broader than the team-production setting that originally motivated it.

Modigliani and Miller (1958, 1963). Shared idea: both papers are theories of what determines firm value as a function of financial structure. Relationship: direct contrast — Modigliani-Miller derive irrelevance (or, with taxes, monotonic debt preference); Jensen and Meckling derive relevance through agency costs, explicitly to resolve the empirical tension the tax-adjusted Modigliani-Miller theory could not. It matters because it is the paper's most direct theoretical foil, engaged with explicitly and at length in Section 3.

DEVELOPED SIMILAR IDEAS CONCURRENTLY

Ross (1973), “The Economic Theory of Agency: The Principal's Problem.” Shared idea: formal principal-agent contracting under divergent interests. Relationship: Ross's treatment is normative (how should the principal design the compensation contract), while Jensen and Meckling's is positive (what contract, and what resulting ownership structure, do self-interested parties actually reach). It matters because the two papers, appearing in the same period, mark the split between the normative mechanism-design branch and the positive corporate-finance branch of agency theory.

APPLIED THE OPTION-PRICING MAPPING

Black and Scholes (1973); Merton (1974); Galai and Masulis (1976). Shared idea: corporate securities as contingent claims on firm value. Relationship: Jensen and Meckling borrow the Black-Scholes mapping of equity to a call option directly to derive the risk-shifting result; Galai and Masulis subsequently apply the same option framework to mergers and investment decisions in a manner that extends the risk-shifting logic developed here. It matters because it shows the paper's debt-agency analysis is not free-standing but embedded in, and dependent on, the contemporaneous option-pricing revolution.

EXTENDED AND CRITIQUED IT

Myers (1977), “Determinants of Corporate Borrowing.” Shared idea: debt distorts investment incentives. Relationship: extends the analysis to a multi-period setting with future growth options, showing debt overhang can cause under-investment in positive-NPV projects through a mechanism distinct from, but observationally similar to, the risk-shifting channel here. It matters because it identifies a second, competing explanation for the same qualitative prediction, as discussed in Section 8 above.

Kürsten (1995), “Risky Debt, Managerial Ownership and Capital Structure: New Fundamental Doubts on the Classical Agency Approach.” Shared idea: direct critique of the Section 5 synthesis. Relationship: shows the additive separability of the debt- and equity-agency-cost schedules assumed in the synthesis does not hold generally once the interaction between managerial ownership and risky debt is modeled jointly, undermining the interior-optimum leverage prediction. It matters because it is the most theoretically serious formal challenge to the paper's headline capital-structure result identified in this review.

Jensen (1986), “Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers.” Shared idea: managerial discretion over firm resources generates agency costs. Relationship: direct extension by one of the original authors to the case of internally generated free cash flow in mature, slow-growth firms with no need for new outside financing, a case the 1976 framework (built around the decision to raise new capital) does not address. It matters because it is the paper's own author's later acknowledgment that the 1976 model's scope was incomplete.

Myers and Majluf (1984), “Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have.” Shared idea: financing choice reflects a friction between managers and outside capital markets. Relationship: substitutes an adverse-selection/information-asymmetry mechanism for the moral-hazard/agency mechanism as the driver of the pecking order of financing choices, generating overlapping but mechanistically distinct predictions about the preference for debt over new equity. It matters because it is the dominant rival explanation, in the modern corporate-finance canon, for several of the same stylized facts Jensen and Meckling's theory targets.

10 Research Gaps

10.1 Paper-Level Gaps

Gap. No formal proof of existence or uniqueness of the interior optimum in the Section 5 synthesis under general (non-additively-separable) cost functions. **Why it matters.** The paper's most cited capital-structure prediction rests on an assumed U-shape that is not derived from primitives. **Existing limitation.** Kürsten (1995) shows the assumption fails in a natural generalization. **Possible way forward.** A joint optimization over $(\alpha, D/(D+S))$ with the risk-shifting incentive parameterized as a function of both variables simultaneously, solved for existence and comparative statics.

Gap. No treatment of multi-period reputation effects. **Why it matters.** Bonding costs in practice are plausibly lower for repeat issuers with reputational capital at stake, which the single-decision framework cannot capture. **Existing limitation.** Assumption P.6 rules this out by construction. **Possible way forward.** A repeated-game extension in which the manager's bonding expenditure choice signals type across financing rounds.

10.2 Literature-Level Gaps

Gap. The relative empirical importance of the agency-cost channel versus the information-asymmetry channel (Myers and Majluf, 1984) in explaining observed financing hierarchies remains unresolved across the literature. **Why it matters.** The two theories often predict the same sign (preference for internal funds and debt over new equity) but for different reasons, and policy implications (disclosure regulation versus governance regulation) differ. **Existing limitation.** Cleanly separating identification of the two channels in observational data is difficult because both correlate with firm opacity and size. **Possible way forward.** Natural experiments that shift information asymmetry (e.g., changes in disclosure regulation) while holding managerial ownership incentives fixed, or vice versa.

10.3 Methodological Gaps

Gap. No calibration of the model's implied magnitude of agency costs against directly measured proxies (perquisite consumption, monitoring expenditure, bond covenant intensity). **Why it matters.** Without magnitude, the theory cannot be ranked in importance against competing frictions (taxes, information asymmetry, transaction costs). **Existing limitation.** Perquisite consumption and monitoring intensity are difficult to observe directly and were largely unmeasured in 1976; modern proxies (executive perquisite disclosure under SEC rules, ISS governance scores, covenant-intensity indices such as Bradley and Roberts, 2015) now make a structural calibration feasible in a way it was not at the time of publication.

11 Research Opportunities

Idea 1 (Extension). Testing the additive-separability assumption of the Section 5 synthesis using modern leveraged-buyout and going-private data

RESEARCH QUESTION Does the marginal agency cost of debt vary with managerial ownership concentration, contradicting the additive-separability assumption Jensen and Meckling's synthesis requires?

MOTIVATION Kürsten's (1995) theoretical critique has, to the reviewer's knowledge, not been comprehensively tested against post-2000 buyout data with detailed sponsor ownership and covenant records.

HYPOTHESIS The sensitivity of covenant intensity (a monitoring-cost proxy) to leverage is itself a function of sponsor equity retained, contrary to independence.

ECONOMIC MECHANISM Risk-shifting incentives depend jointly on both the manager's residual claim and the debt-to-value ratio; the two cannot be varied independently in the manager's true optimization problem.

LITERATURE CONNECTION Kürsten (1995); Bradley and Roberts (2015) on covenant intensity.

DATA Refinitiv LPC DealScan or S&P LCD leveraged loan covenant data merged with sponsor equity stakes from SEC Form D / going-private proxy filings, 2000–2025.

VARIABLES Covenant-intensity index (dependent); leverage ratio, sponsor equity share, and their interaction (independent); industry and year fixed effects (controls).

METHOD Panel regression with an explicit leverage $\times$ ownership interaction term; test whether the interaction coefficient is significantly different from zero.

IDENTIFICATION STRATEGY Exploit staggered private-equity fund vintage effects and sponsor-level fixed effects that shift target ownership concentration for reasons plausibly unrelated to the specific target's risk-shifting opportunity set.

EXPECTED CONTRIBUTION A direct empirical test of a 50-year-old, widely assumed but formally disputed separability condition underlying capital-structure theory.

MAIN RISK Covenant intensity is a noisy and multidimensional proxy for monitoring cost; the interaction effect may be confounded by unobserved deal-quality selection.

Idea 2 (New Application). Agency costs of outside equity in founder-controlled dual-class technology firms

RESEARCH QUESTION Does the under-investment prediction of equation (4) hold among founder-CEOs who retain supervoting shares but sell large cash-flow (non-voting economic) stakes, a structure Jensen and Meckling's non-voting-equity assumption (P.3) approximates closely?

MOTIVATION Dual-class technology IPOs since roughly 2010 create a natural laboratory in which economic ownership and control are separated in exactly the way assumption P.3 stipulates, unlike the diversified single-class ownership typical of most public equity.

HYPOTHESIS Capital expenditure and R&D investment, scaled by assets, are lower relative to a value-maximizing benchmark (proxied by analyst NPV-positive project disclosures or peer investment rates) as founder cash-flow rights fall, holding voting control fixed.

ECONOMIC MECHANISM The Section 3.2 under-investment result: as α (here, cash-flow rights) falls, the founder captures a smaller share of the value created by new investment relative to the private cost (effort, non-pecuniary loss) of pursuing it.

LITERATURE CONNECTION Direct test of Section 3.2 of this paper; relates to the modern dual-class-discount literature (Smart, Thirumalai, and Zutter, 2008).

DATA Compustat capital expenditure and R&D data merged with dual-class share structure and founder cash-flow-rights disclosures from proxy statements, U.S. technology IPOs 2010–2024.

VARIABLES Investment rate (dependent); founder cash-flow-rights percentage (independent, varying over time as founders sell down); voting-rights percentage (control, to isolate cash-flow from control effects).

METHOD Firm fixed-effects panel regression exploiting within-founder variation in cash-flow rights over time as lockups expire and secondary sales occur.

IDENTIFICATION STRATEGY Lockup-expiration-driven step changes in cash-flow rights provide quasi-exogenous timing variation, holding voting control constant by construction of the dual-class structure.

EXPECTED CONTRIBUTION A clean, modern test of the paper's core under-investment mechanism in a setting where cash-flow and control rights are observably separated, which was not possible with 1976-era ownership data.

MAIN RISK Founders may sell down cash-flow rights precisely when firm-specific investment opportunities are shrinking, creating reverse causality that the lockup-timing instrument only partially addresses.

Idea 3 (Replication with Modern Data). Re-examining the regulated-industry leverage prediction

RESEARCH QUESTION Do U.S. regulated utilities and banks still exhibit higher debt-equity ratios than unregulated firms of comparable systematic risk, as the paper's ninth stated implication predicts, after five decades of financial deregulation?

MOTIVATION The claim was asserted narratively in 1976 without a supporting data series; it has never, to the reviewer's knowledge, been directly re-tested against the full post-deregulation sample.

HYPOTHESIS The regulated-versus-unregulated leverage gap has narrowed since major deregulation episodes (e.g., the Riegle-Neal Act of 1994, state-level utility restructuring in the 1990s), consistent with the theory if regulatory monitoring substitutes for bondholder monitoring.

ECONOMIC MECHANISM Regulatory oversight lowers the marginal agency cost of debt at a given leverage ratio by substituting for costly private monitoring; deregulation should raise that marginal cost back toward the unregulated-firm level.

LITERATURE CONNECTION Direct test of the paper's ninth introductory claim; relates to the regulation-and-capital-structure literature.

DATA Compustat leverage ratios by industry (SIC/NAICS regulated-utility and banking codes) and risk (asset beta), 1970–2025, spanning multiple regulatory regimes.

VARIABLES Book and market leverage (dependent); regulated-industry indicator, systematic risk, regulatory-regime indicator, and their interaction (independent).

METHOD Difference-in-differences around discrete deregulation events, controlling for asset risk and firm size.

IDENTIFICATION STRATEGY Staggered timing of state-level utility deregulation provides quasi-experimental variation in the intensity of regulatory monitoring, holding the underlying business largely fixed.

EXPECTED CONTRIBUTION A long-overdue direct empirical test, with a fifty-year data series unavailable to the original authors, of one of the paper's explicit, falsifiable claims.

MAIN RISK Deregulation events are often correlated with broader macroeconomic and interest-rate shifts that independently affect leverage across all industries, threatening the difference-in-differences identification.

Idea 4 (New Methodology). Structural estimation of the joint agency-cost-minimization problem

RESEARCH QUESTION Can the parameters of the manager's utility function over wealth and perquisites, and the monitoring-cost function $F(M, \alpha)$, be structurally estimated from observed ownership, leverage, and covenant data, and does the estimated model rationalize the observed cross-sectional distribution of capital structures?

MOTIVATION The paper is purely diagrammatic; no one, to the reviewer's knowledge, has structurally estimated its primitives directly, as opposed to testing reduced-form implications.

HYPOTHESIS A structural model with plausible curvature in the monitoring-cost function can match the observed joint distribution of managerial ownership stakes and leverage ratios in the Compustat/ExecuComp universe.

ECONOMIC MECHANISM Direct structural implementation of equations (1)–(6) above.

LITERATURE CONNECTION Methodologically related to structural corporate-finance estimation (e.g., Hennessy and Whited, 2005, on structural investment-financing models).

DATA ExecuComp managerial ownership, Compustat leverage and investment data, DealScan covenant data, U.S. public firms.

VARIABLES Managerial ownership share, leverage, investment rate, covenant intensity, jointly modeled as endogenous outcomes of the structural optimization.

METHOD Simulated method of moments or maximum likelihood structural estimation.

IDENTIFICATION STRATEGY Cross-sectional and time-series moments of the joint ownership-leverage-investment distribution identify the curvature parameters of the cost functions under standard structural-estimation regularity conditions.

EXPECTED CONTRIBUTION Would convert a fifty-year-old qualitative theory into a quantitatively disciplined structural model, directly addressing the methodological gap identified in Section 10.3.

MAIN RISK Structural identification of the monitoring-cost function separately from competing frictions (taxes, information asymmetry) is demanding and may require restrictive functional-form assumptions that themselves become the object of critique.

Idea 5 (New Market / Contradiction Test). Agency costs of "outside equity" in over-collateralized DeFi lending protocols

RESEARCH QUESTION Does the risk-shifting mechanism of Section 3.3 manifest in decentralized-finance lending protocols, where "management" (protocol governance token holders) can vote to change collateral and liquidation parameters after loans are extended?

MOTIVATION DeFi protocols create a setting with pseudonymous, algorithmically enforced debt contracts and observable on-chain governance votes, offering unusually clean data on a modern analogue of the bondholder-manager conflict the paper models.

HYPOTHESIS Governance votes that loosen collateral or risk parameters are systematically associated with a rise in the protocol's native governance-token price (an equity-like claim) and a fall in the implied value of outstanding lending positions (a debt-like claim), mirroring equation (6).

ECONOMIC MECHANISM Governance-token holders, as residual claimants on protocol fees, have the same incentive to shift risk toward fixed-claim lenders that the paper's equity holders have toward bondholders, but with the advantage that both the "vote" (investment decision) and the resulting price changes are fully observable on-chain.

LITERATURE CONNECTION Direct application of Section 3.3 of this paper to a genuinely new asset class not contemplated by the original authors.

DATA On-chain governance proposal and vote records, token prices, and protocol total-value-locked data from major lending protocols (e.g., Aave, Compound), 2020–2026.

VARIABLES Governance-token abnormal return and lending-position implied value change (dependent) around risk-parameter votes; vote outcome and magnitude of parameter change (independent).

METHOD Event study around governance vote resolution dates.

IDENTIFICATION STRATEGY Close governance votes (passing or failing by a narrow margin) provide a regression-discontinuity-style source of quasi-random variation in whether the risk-shifting parameter change is implemented.

EXPECTED CONTRIBUTION Extends a fifty-year-old theoretical mechanism to a genuinely novel institutional setting with unusually clean, fully observable "contracting" data, and speaks to ongoing debates about DeFi governance design.

MAIN RISK Governance votes are infrequent and often confounded with simultaneous protocol news, and the "bondholder" (lender) side of DeFi protocols typically also receives variable, risk-compensating interest rates, which may already price the risk-shifting possibility and mute the predicted transfer.

12 Research Idea Ranking

Ranked by a joint assessment of novelty, feasibility, data availability, methodological quality, economic importance, and potential contribution, the top three ideas are Idea 1, Idea 3, and Idea 2, in that order.

Idea 1 ranks highest because it directly targets the single weakest link identified in the critical evaluation — the additive-separability assumption underlying the paper's headline capital-structure prediction — using data (detailed covenant and sponsor-ownership records) that did not exist in any usable form until the growth of the leveraged-loan and private-equity data industry in the 2000s; feasibility is high because DealScan-style covenant data is a well-established resource in the literature.

Idea 3 ranks second because it is the most directly falsifiable test of an explicit claim the authors themselves state in the paper, uses widely available Compustat data, and exploits genuine quasi-experimental variation from documented deregulation events, making it highly feasible with moderate but real identification risk from confounding macro shocks.

Idea 2 ranks third because the dual-class technology-IPO setting offers an unusually clean separation of cash-flow and control rights that closely matches the paper's own assumption structure, but the sample of qualifying firms is smaller and the lockup-expiration identification strategy is less sharp than the discontinuities exploited in Ideas 1 and 3.

Idea 4 is ranked below the top three principally on feasibility grounds: structural identification of the monitoring-cost function separately from competing frictions is demanding and may not be robust to functional-form choices. Idea 5, while the most novel, carries the highest execution risk because DeFi governance votes are infrequent, often confounded with contemporaneous protocol news, and because lending-rate adjustments may already partially price the risk-shifting possibility, muting the predicted effect.

13 Learning Framework

13.1 Prerequisite Concepts

Constrained utility maximization and indifference-curve analysis; the Sharpe-Lintner Capital Asset Pricing Model, sufficient to understand why two log-normal payoff distributions with equal expected value and equal systematic risk have equal market value; the Black-Scholes-Merton option-pricing framework, sufficient to understand equity as a call option on firm value; and the Modigliani-Miller capital-structure irrelevance propositions, both the untaxed and tax-adjusted versions, as the theoretical benchmark this paper departs from.

13.2 Core Concepts Learned

The tripartite decomposition of agency costs into monitoring expenditures, bonding expenditures, and residual loss; the pricing theorem showing that anticipated agency costs are capitalized into the price a seller receives, so the seller bears them even though a third party might appear to be the one "monitoring"; the distinction between the agency costs of outside equity (perquisite consumption, under-investment) and the agency costs of debt (risk-shifting, monitoring, bonding, bankruptcy costs); and the framing of ownership structure as the outcome of minimizing the sum of these costs rather than as an exogenously given institutional fact.

13.3 Important Equations

Equations (1) through (6) in Section 3 above, in particular the pricing-theorem wealth identity (equation 2), the under-investment condition (equation 4), and the risk-shifting value decomposition (equation 6), are the equations worth being able to reproduce and explain economically without reference to this review.

13.4 Methods Learned

Diagrammatic comparative-statics analysis using indifference curves and budget-constraint-like "opportunity loci"; proof by contradiction as applied to a competitive-pricing equilibrium argument; and the mapping of corporate securities onto option payoffs as a tool for signing incentive effects.

13.5 Questions to Test Understanding

Comprehension. In your own words, why does the owner-manager, and not the outside buyer, bear the full cost of the manager's anticipated increase in perquisite consumption after an equity sale?

Analysis. Explain why the paper's under-investment result (equation 4) requires the net-value function $V(I)-I$ to be concave, and describe what would happen to the sign of the result if the function were instead convex over the relevant range.

Research. Design a test that could, in principle, distinguish the Jensen-Meckling perquisite-driven under-investment channel from the Myers (1977) debt-overhang under-investment channel in a sample of levered firms with new growth options.

13.6 Papers to Read Next

Myers (1977), "Determinants of Corporate Borrowing"; Myers and Majluf (1984), "Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have"; and Jensen (1986), "Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers."

14 Conclusion

One-Sentence Thesis

The ownership structure of a firm is the outcome of a value-maximizing entrepreneur-manager minimizing the sum of the agency costs generated by selling outside equity (perquisite consumption and under-investment) and by selling debt (risk-shifting, monitoring, bonding, and bankruptcy costs), given his personal wealth constraint and the amount of outside financing the firm requires.

Three Most Important Insights

First, agency costs are capitalized into the price a manager receives for the claims he sells, so he alone bears them and therefore has a first-best incentive to minimize them, even though he cannot eliminate them. Second, outside equity and debt generate agency costs through entirely different, formally distinct channels — perquisite consumption versus risk-shifting — so the optimal ownership structure is a genuine tradeoff between two qualitatively different frictions, not a single-dimensional problem. Third, a reduction in firm value caused by positive agency costs is not evidence of inefficiency relative to any attainable alternative; it is evidence of the unavoidable cost of raising capital from anyone other than a sole, unconstrained owner.

Strongest Evidence

The pricing theorem of Section 3.2, proved rigorously by contradiction within the paper itself, is the strongest single result: it does not depend on auxiliary assumptions about functional form and establishes unambiguously that the manager bears the full wealth consequence of the ownership structure he chooses.

Biggest Weakness

The additive separability of the debt- and equity-agency-cost schedules assumed in the Section 5 synthesis, which subsequent theoretical work (Kürsten, 1995) shows does not hold once the interaction between managerial ownership and risky debt is modeled jointly, weakening the paper's best-known specific prediction — a well-defined interior optimal leverage ratio — relative to its more robust qualitative insights about the existence and direction of the two agency-cost channels.

Most Important Research Gap

The absence, fifty years on, of a structural estimation of the model's primitives against the joint cross-sectional distribution of managerial ownership, leverage, and covenant intensity, which would finally allow the theory's qualitative mechanisms to be ranked by economic magnitude against competing explanations such as information asymmetry and taxation.

Best Research Opportunity

Idea 1 above — testing the additive-separability assumption directly against modern leveraged-buyout covenant and sponsor-ownership data — because it is the most direct, most feasible test of the specific theoretical weakness identified as most consequential in this review.

Three Papers to Read Next

Myers (1977); Myers and Majluf (1984); Jensen (1986).

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SOURCE ACCESS NOTE

Primary source: Jensen, M. C., and Meckling, W. H. (1976), "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure," Journal of Financial Economics 3(4), 305–360, DOI 10.1016/0304-405X(76)90026-X, confirmed openly accessible without paywall via SSRN (papers.ssrn.com/sol3/papers.cfm?abstract_id=94043) and retrieved in full-text form for the purposes of this review from a university course-reserve mirror (josephmahoney.web.illinois.edu). Verbatim primary-source text was retrieved and directly used for Sections 1 through 4.3 of the original article (pages 305–341 of 360). The synthesis in the original article's Section 5 and the qualifications in its Section 6 could not be retrieved verbatim within this session; the characterization of those sections in this review (principally Section 3.5 above) is reconstructed from the paper's stated introduction, from the internal logic of Sections 2 and 4, and cross-checked against an independent peer-reviewed secondary summary (Gao, 2019) rather than quoted directly from the original pages. This limitation is disclosed here in accordance with the standing instruction to state plainly when material cannot be established from directly available sources. A machine-readable copy of the paper (paper.pdf) could not be saved to this review's library folder because the sandboxed research environment used to prepare this review restricts outbound network access to a fixed domain allowlist that does not include the hosts serving this paper; the source has nonetheless been confirmed openly accessible to a normal browser at the addresses given above.